

Research in **Data** **Science**

Benchmarking EU Financial Regulatory QnA with
LLM-as-a-Judge Evaluation

Datasets

The dataset consists of **2,669 EBA** and **399 ESMA** question-answer pairs scraped directly from official **EBA** and **ESMA** websites, representing real regulatory Q&A on banking and securities.

Why this dataset?

Authoritative Source

Sourced directly from official publications ensuring ground truth.

Broad Domain Coverage

Covers EU financial regulatory topics across banking and securities.

Novel Benchmark

The first benchmark specifically focused on EU financial regulatory QnA.

Example of EBA Question

```
"id": "2015_1856",
"status": "Final Q&A",
"legal_act": "Regulation (EU) No 575/2013 (CRR)",
"topic": "Supervisory reporting - Leverage ratio",
"subject_matter": "Other Asset",
"question": "Should all the accounting based off-settings (here including for example tax assets/ liabilities off-setting) be grossed-up on Other Asset of LR exposure?",
"background": "Annex XI states that\n\"All assets other than contracts listed in Annex II of the CRR, credit derivatives, repurchase transactions, securities or commodities lending or borrowing transactions, long settlement transactions and margin lending transactions. Institutions shall base valuation on the principles set out in Article 429 (5) of the CRR.\"\nAccording to Article 429(5) CRR:\n(a) the exposure values of assets means exposure values in accordance with the first sentence of Article 111(1)\n(c) loans shall not be netted with deposits\".\nAccording to Article 111 CRR \"The exposure value of an asset item shall be its accounting value remaining after specific credit risk adjustments, additional value adjustments in accordance with Articles 34 and 110 and other own funds reductions related to the asset item have been applied.\",
"final_answer": "Article 429(5)(a) of Regulation (EU) No 575/2013 (CRR), as amended by the Delegated Regulation (EU)2015/62 (DR), specifies that the leverage ratio exposure value of assets, including SFTs, should be calculated in accordance with the first sentence of Article 111(1) CRR, pursuant to which the exposure value of an asset item shall be its accounting value remaining after specific credit risk adjustments, additional value adjustments in accordance with Articles 34 and 110 and other own funds reductions related to the asset item have been applied.\nArt. 195 CRR also clarifies that \"an institutions may use on-balance sheet netting of mutual claims between itself and its counterparty as an eligible form of credit risk mitigation\" (CRM), thus including the on-balance sheet netting of mutual claims among the forms of CRM.\nGiven that the instructions for the position \"Other assets\" in Annex XI of the Regulation (EU) No 680/2014 (ITS on Supervisory Reporting) also refer specifically to the principles set out in Article 429(5) of the CRR, which allow no netting nor application of CRM, offsetting relating to tax assets / liabilities should be reversed.",
"published_date": "04/10/2019",
"answer_source": "Answer prepared by the EBA.",
"url": "https://www.eba.europa.eu/single-rule-book-qa/qna/view/publicId/2015_1856"
```

Example of ESMA Question

"id": "ESMA_QA_2557",

"status": "Final",

"level1_regulation": "Markets in Financial Instruments Regulation (MiFIR) Regulation (EU) No 600/2014 - Investor Protection and Intermediaries",

"topic": "Inducements",

"subject_matter": "Investment firms' use of rebates and discounts to lower fees for clients under the payment for order flow (PFOF) prohibition",

"question": "Under Article 39a of Regulation (EU) 2014/600/EU (MIFIR), are investment firms allowed to use rebates and discounts on the transaction fees of execution venues to lower fees for clients?",

"answer": "Answer provided by the European Commission\n\nYes, investment firms are allowed to use rebates and discounts to lower fees for clients but only where they comply and are able to demonstrate their compliance with all of the following conditions set out in Article 39a(1) of MiFIR:\n(a) the rebate or discount on the transaction fees of execution venues is permitted under the approved and public tariff structure of the relevant trading venue in the Union or of the relevant third-country trading venue;\n(b) the rebate or the discount exclusively benefits the client and not the firm;\n(c) they do not result in a monetary benefit to the investment firm\n\nDisclaimer:\n\nThe answers clarify provisions already contained in the applicable legislation. They do not extend in any way the rights and obligations deriving from such legislation nor do they introduce any additional requirements for the concerned operators and competent authorities. The answers are merely intended to assist natural or legal persons, including competent authorities and Union institutions and bodies in clarifying the application or implementation of the relevant legal provisions. Only the Court of Justice of the European Union is competent to authoritatively interpret Union law. The views expressed in the internal Commission Decision cannot prejudice the position that the European Commission might take before the Union and national courts.",

"response_date": "03-03-2026",

"published_date": "03/06/2025",

"url": "https://www.esma.europa.eu/publications-data/questions-answers/2557"

LLM as a-Judge Prompt : Answerer

*You are an expert in EU financial regulation, specializing in EBA and ESMA guidelines.
Answer the following regulatory question accurately and completely.
Where relevant, reference specific articles, guidelines, or regulatory frameworks.*

LLM as a-Judge Prompt : Judge

You are an expert judge evaluating answers to EU financial regulation questions \ from EBA and ESMA sources.

Question

{question}

Official Answer (Ground Truth)

{ground_truth}

Candidate Answer

{candidate}

Scoring Rubric

Score each dimension 1-5:

- 1: Completely wrong/missing*
- 3: Partially correct/complete*
- 5: Fully correct/complete*

Dimensions:

- Accuracy: factual correctness vs official answer*
- Completeness: coverage of all key points*
- Clarity: clear, structured, unambiguous language*
- Overall: weighted average (accuracy 50%, completeness 30%, clarity 20%)*

Instructions

- 1. Write your reasoning FIRST*
- 2. Then assign scores based on your reasoning*

Respond in this exact JSON format only:

```
{{  
  "reasoning": "<two sentences evaluating the answer>",  
  "accuracy": "<1-5>,"  
  "completeness": "<1-5>,"  
  "clarity": "<1-5>,"  
  "overall": "<1-5>"  
}}""
```

Models

Small Models :

*--answerer Llama-3.1-8B-Instruct *

*--judge Qwen3.5-9B *

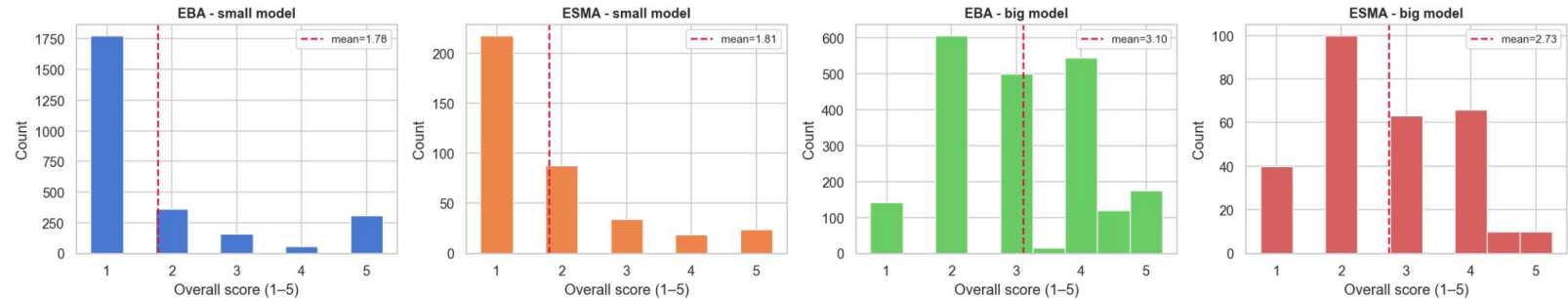
Bigger Models :

--answerer "swiss-ai/Apertus-70B-Instruct-2509"

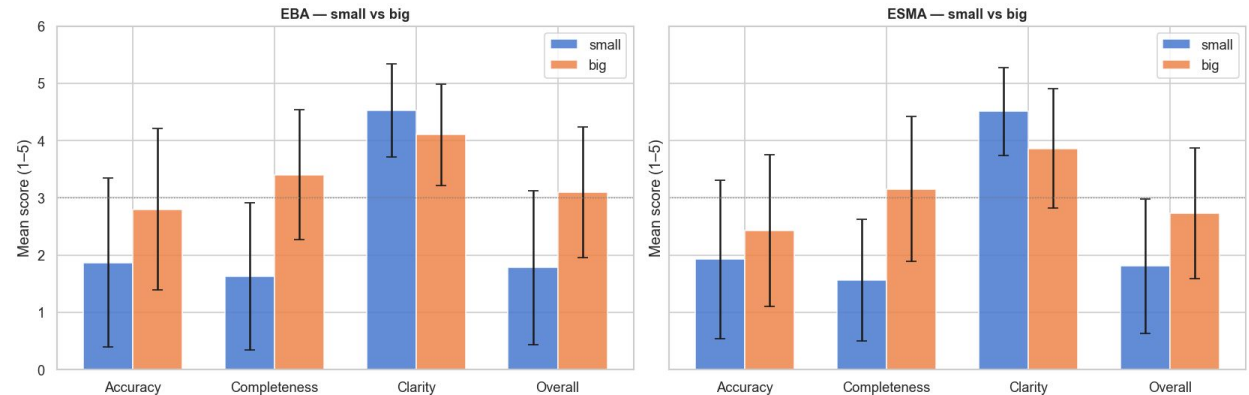
--judge "meta-llama/Llama-3.3-70B-Instruct"

Judge Results - Overall Distribution

Overall Score Distribution



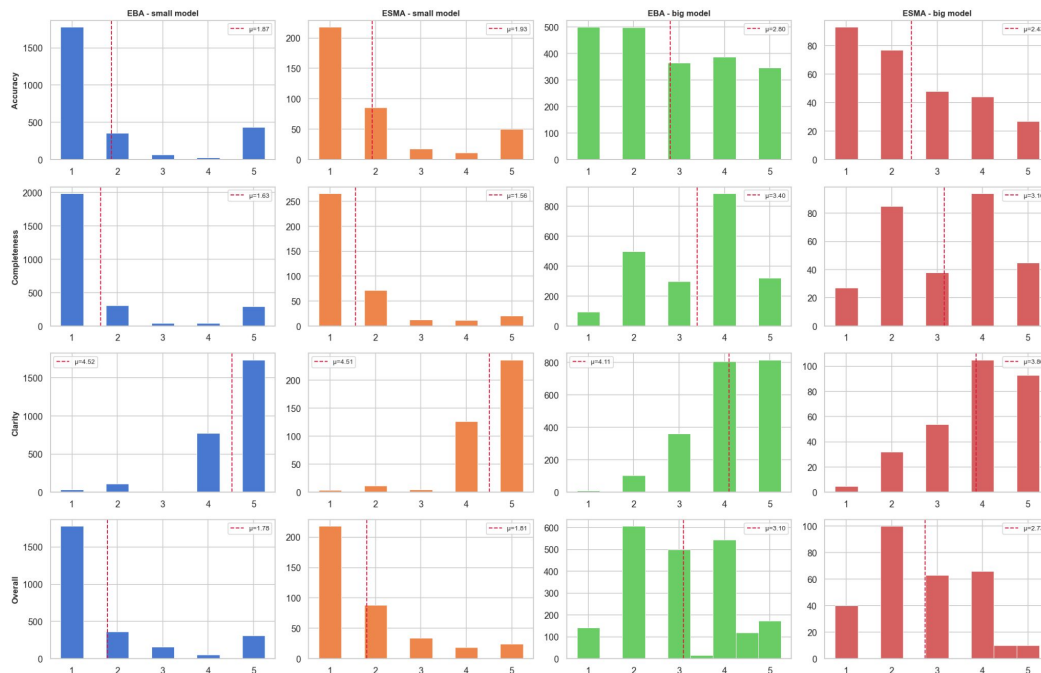
Mean Scores by Dimension: Small vs Big Models



Small models score very low (mean ≈ 1.8 , heavily piled at 1), big models score much higher (mean ≈ 3.1 EBA / 2.7 ESMA) with a more spread distribution.

Judge Results - Score in Each Dimensions

Score Distributions — All Dimensions



The small models seems off :

- Gives small score for accuracy and completeness
- While giving a high score for the clarity

While for big models, the distributions are more fair.

Next step :

- Take a look at Cohen's weighted kappa disagreement rate and krippendorff alpha to measure agreement among any number of raters

Judge Results - Disagreement Rate (Judge 1 and Judge 2)

Cohen's Weighted Kappa

$$\kappa_w = 1 - \frac{\sum_{i=1}^k \sum_{j=1}^k w_{ij} o_{ij}}{\sum_{i=1}^k \sum_{j=1}^k w_{ij} e_{ij}}$$

Linear Weight Schemes

$$w_{ij} = \frac{|i - j|}{k - 1}$$

Quadratic Weight Schemes

$$w_{ij} = \frac{(i - j)^2}{(k - 1)^2}$$

- Where:
- k = number of categories (5 in your case)
- o_{ij} = observed proportion of items where rater 1 gave i and rater 2 gave j (cell of normalized confusion matrix)
- $e_{ij} = p_i \cdot p_j$ expected proportion under independence (product of marginals)
- w_{ij} = disagreement weight between categories i and j

Krippendorff's alpha

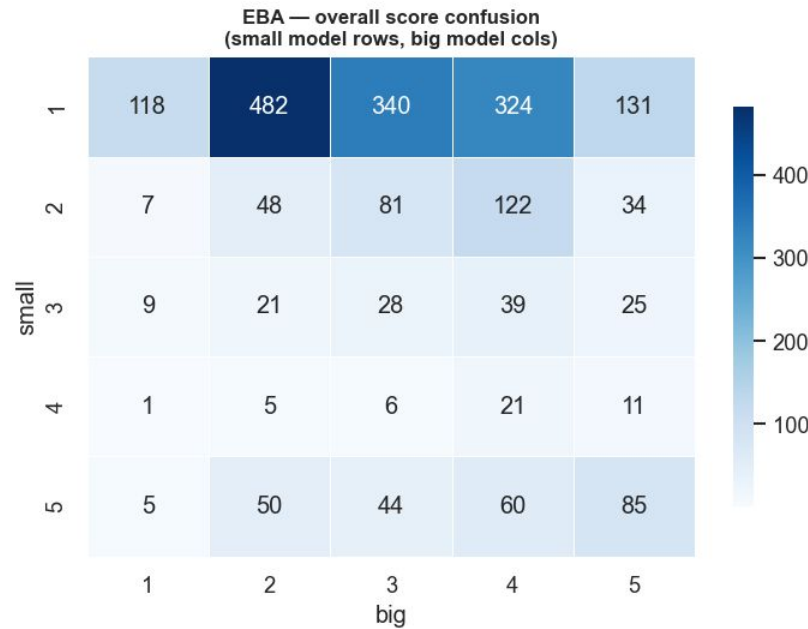
$$\alpha = 1 - (n - 1) \cdot \frac{\sum_c \sum_k o_{ck} \delta_{ck}^2}{\sum_c \sum_k n_c n_k \delta_{ck}^2}$$

Where:

- o_{ck} = observed coincidences (pairs of ratings c and k on the same item)
- n_c, n_k = total occurrences of values cc and kk
- n = total number of pairable values
- δ_{ck}^2 = squared difference function, $\delta^2 = (c - k)^2$ for interval measurement

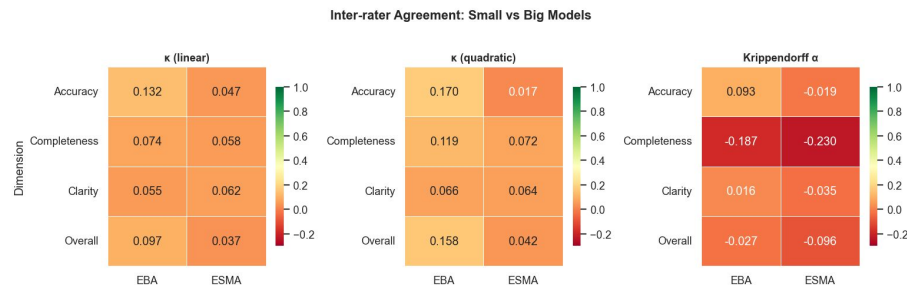
Judge Results - Disagreement Rate (Judge 1 and Judge 2)

Confusion Matrix: Small vs Big Overall Scores

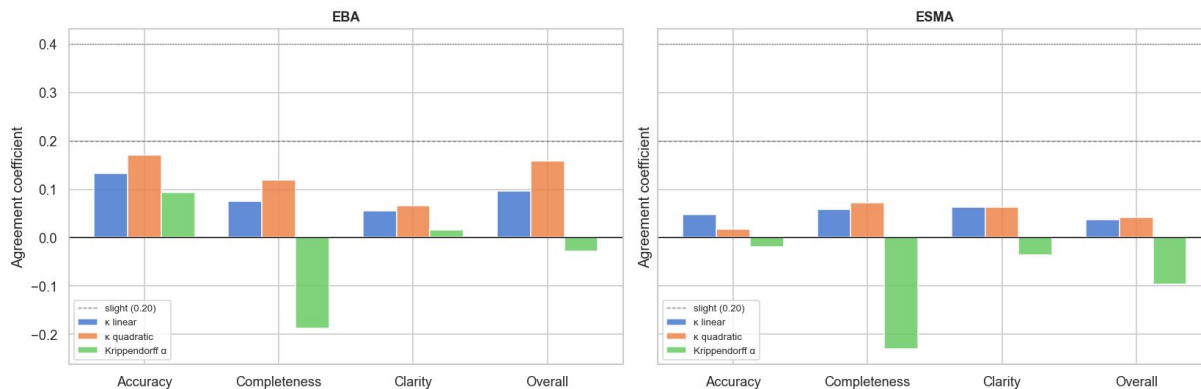


Judge Results - Disagreement Rate (Judge 1 and Judge 2)

corpus	dimension	n_pairs	kappa_linear	kappa_quadratic	krippendorff_alpha
EBA	accuracy	2097	0.132	0.170	0.093
	completeness	2097	0.074	0.119	-0.187
	clarity	2097	0.055	0.066	0.016
	overall	2097	0.097	0.158	-0.027
ESMA	accuracy	289	0.047	0.017	-0.019
	completeness	289	0.058	0.072	-0.230
	clarity	289	0.062	0.064	-0.035
	overall	289	0.037	0.042	-0.096



Cohen's κ and Krippendorff α per Dimension



The disagreement rate is high, which means the small model produces punished score than the big model. The highest kappa rate is in accuracy dimensions on EBA, which is still very small to conclude that they have agreement across the models

Future work :

- Try take a look at descriptive results of each example of judges with the biggest rate disagreement

Judge Results

Example of Judges Results with Biggest Disagreement Rate

Example 1 :

JUDGE REASONING (small): The candidate answer is factually incorrect because it directly contradicts the official answer by claiming the definitions are not identical, whereas the regulation explicitly states they are identical.

JUDGE REASONING (big): The candidate answer accurately reflects the official answer by confirming that the definition of exposure for Large Exposure Reporting is the same as in the Standardised Approach, including the balance sheet value with accrued interest and impairments deducted, and provides additional supporting information from EBA guidelines. The answer is clear, well-structured, and covers all key points, demonstrating a thorough understanding of the topic and relevant regulatory frameworks.

small={'accuracy': 1.0, 'completeness': 1.0, 'clarity': 5.0, 'overall': 1.0}

big={'accuracy': 5.0, 'completeness': 5.0, 'clarity': 5.0, 'overall': 5.0}

Judge Results

Example of Judges Results with Biggest Disagreement Rate

Example 2 :

JUDGE REASONING (small): The candidate answer perfectly aligns with the official ground truth by correctly specifying that positions must be aggregated in EUR equivalent using spot exchange rates at the time of the transaction, regardless of currency bucket differences.

JUDGE REASONING (big): The candidate answer fails to accurately reflect the official answer's requirement for trade repositories to calculate and express aggregate positions in SFTs in their EUR equivalent value, instead providing a misleading and overly complicated explanation of aggregation methodologies. The answer lacks clarity and completeness, as it does not directly address the question and introduces unnecessary references to other guidelines, resulting in a low overall score.

```
small={'accuracy': 5.0, 'completeness': 5.0, 'clarity': 5.0, 'overall': 5.0}
```

```
big={'accuracy': 1.0, 'completeness': 2.0, 'clarity': 2.0, 'overall': 1.0}
```

Judge Results

Example of Judges Results with Biggest Disagreement Rate

Future work :

- *Run the judges again with probability to have high agreement rate*
- *Compare with RAG-Models judges results*
- *Probes where the models is failing*